Homework 6.3 #1

Student Name: ______________________

Class Name: **College Algebra FA26 - 014 MW 2:30**

Number of Questions: **5**

Instructor Name: **Mamani Velasco, Ruben**

Question 1 of 5

Suppose that the augmented matrix of a system of linear equations for unknowns x, y, and z is

$$\left[\begin{array}{ccc|c} 1 & 0 & -2 & 1 \\ 5 & 1 & -14 & 10 \\ \frac{10}{3} & 0 & -\frac{20}{3} & \frac{10}{3} \end{array}\right].$$

Solve the system and provide the information requested.
The system has:

____ a unique solution, which is

x=_____ y=_____ z=_____

____ infinitely many solutions, two of which are

x=_____ y=_____ z=_____

x=_____ y=_____ z=_____

____ no solution

Question 2 of 5

Let $A=\begin{bmatrix} 2 & -5 \\ -6 & -3 \\ 7 & 4 \end{bmatrix}$. Find $2A$.

Question 3 of 5

Let $B = \begin{bmatrix} 7 & 1 \\ 5 & -6 \end{bmatrix}$ and $C = \begin{bmatrix} 4 & 0 \\ -5 & 3 \end{bmatrix}$.

Find $B - C$.

Question 4 of 5

Let $F = \begin{bmatrix} 0 & 3 \\ 7 & -5 \end{bmatrix}$ and $G = \begin{bmatrix} -5 & 2 \\ 1 & 0 \end{bmatrix}$.

Find $-4F - 3G$.

Question 5 of 5

Amanda, Manuel, and Melissa each sell boxes of strawberries for $\$1.70$ each, tomatoes for $\$0.80$ each, and avocados for $\$1.60$ each. They don't sell any other food. The number sold by each seller and each type of food is as follows.

Number sold			
	Amanda	Manuel	Melissa
Boxes of strawberries	400	420	440
Tomatoes	620	590	570
Avocados	530	480	520

The table can be represented as the following matrix.

$$A = \begin{bmatrix} 400 & 420 & 440 \\ 620 & 590 & 570 \\ 530 & 480 & 520 \end{bmatrix}$$

(a) Find a matrix B whose entries are the prices (in dollars) of each type of food. Make sure the product BA is defined and its entries are the amounts of money (in dollars) made by each seller overall.

$B = \$$

(b) Calculate the product BA.

$BA = \$$

(c) How much money (in dollars) did Manuel make overall?

$\$$________

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Question 1 of 5

The system has:

___ infinitely many solutions, two of which are

x=1, y=5, z=0

x=$1+2$, y=$5+4$, z=1

Question 2 of 5

$$2A=\begin{bmatrix} 4 & -10 \\ -12 & -6 \\ 14 & 8 \end{bmatrix}$$

Question 3 of 5

$$B-C=\begin{bmatrix} 3 & 1 \\ 10 & -9 \end{bmatrix}$$

Question 4 of 5

$$-4F-\ 3G=\begin{bmatrix} 15 & -18 \\ -31 & 20 \end{bmatrix}$$

Question 5 of 5

(a) Find a matrix B whose entries are the prices (in dollars) of each type of food. Make sure the product BA is defined and its entries are the amounts of money (in dollars) made by each seller overall.

$$B = \$\begin{bmatrix} 1.70 & 0.80 & 1.60 \end{bmatrix}$$

(b) Calculate the product BA.

$$BA = \$\begin{bmatrix} 2024 & 1954 & 2036 \end{bmatrix}$$

(c) How much money (in dollars) did Manuel make overall?

$\$1954$